

Amarnath V

Robotics Engineer

Results-driven Robotics Engineer with 4 years of experience across R&D, product development, and deployment of robotic systems. Skilled in leading cross-functional projects involving humanoids, AGVs, and robotic manipulators. Strong programming and AI/ML background, with a passion for advancing robotics innovation

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EDUCATION

Japanese Language Course

Miyazaki Information Business College

Japanese Language

10/2025 - Present

Bachelor of Engineering Mechatronics

RVS Technical Campus-Coimbatore/Anna

University Chennai-25

08/2017 - 08/2021,

8.44 CGPA

KEY PROJECTS

iNav (Airforce Project) (08/2025 - 09/2025)

- Developed control and integration software for a towbar tug used in towing aircraft on airfields.
- Enabled efficient and reliable ground-handling operations by implementing robust motion and safety systems.

iTow 20T (Navy Project) (02/2024 - 09/2025)

- Lead programmer for a semi-autonomous towbarless tug designed for Navy helicopters on airfields and ship decks.
- Implemented safety systems, collision-avoidance algorithms, and autonomous navigation features.

Biped Robotic System-1 (DRDO Project) (08/2023 - 08/2025)

- Designed and integrated motion control for a humanoid robot with torso and legs.
- Programmed 14 actuators with INS and F/T sensor feedback to achieve coordinated movements.

Artimus (01/2023 - 04/2023)

- Developed a robotic system for automated cutting of bails in tire manufacturing industries.
- Optimized reliability and safety through custom integration of actuators and sensors.

Autojack 2000 (12/2022 - 04/2023)

- Built an indoor Autonomous Mobile Robot (AMR) for pallet transport in industrial environments.
- Implemented controlling and obstacle-avoidance programs for safe operations.

RoboPack (Gabon, Africa) (08/2022 - 12/2022)

- Programmed robotic manipulator for automated bottle packing into cartons.
- Developed motion planning for simultaneous multi-bottle handling, improving efficiency.

Humanoid Robot using Raspberry pi (10/2021)

- Created a speech-enabled humanoid using Raspberry Pi for robot control and interaction.
- Implemented database-backed speech recognition and coordinated body movements.

Robotic Arm for paralyzed people (10/2021)

- Developed assistive robotic arm using image processing to detect lip movement.
- Programmed Arduino-based servo control to enable feeding operations for paralyzed individuals.

WORK EXPERIENCE

Robotics Engineer

Premier Astra

08/2023 - 09/2025,

Coimbatore, India

Achievements/Tasks

Advanced Protocols and Robotics Integration : Implemented OpenCAN with EtherCAT support for robust communication in robotics projects. Integrated standard components like servo drives, motion controllers, high-end IMU sensors (INS and F/T sensors), and custom actuators across diverse applications.

Motion Control and Software Development : Managed precise control of 14 actuators in a bipedal system using advanced methods. Developed Linux-based motion control program, overcoming library limitations. Leveraged ROS and ROS2 for integrated system functionality and enhancement.

SLAM and Localization : Applied SLAM algorithms and utilized AMCL and GMapping packages for precise mapping and localization in AUGVs.

Jr. Robotics Engineer

Gridbots Technologies Pvt. Ltd.

07/2022 - 05/2023,

Ahmedabad, India

Achievements/Tasks

- **Project Contributions** : Led initiatives involving self-driving robots, robotic manipulators, and fully integrated robotic systems within the Research & Development and Deployment departments.
- **Problem Solving and Innovation** : Demonstrated expertise in resolving challenges through innovative solutions, coupled with proficiency in embedded C programming for Atmega 8, 16, and 32 microcontrollers.
- **Technical Skills Development** : Enhanced embedded systems proficiency and problem-solving abilities through hands-on experience in SPI, I2C, serial, and CAN protocols. Applied knowledge to various industrial sensors and AC/DC actuators, including servo motors and BLDC motors.

Robotics Engineer

Ore Equipe Robotics Research Centre

09/2021 - 07/2022,

Kochi, India

Achievements/Tasks

- **Project Work** : Gained hands-on experience by working on various projects using Raspberry Pi, Arduino, and Node MCU.
- **Training** : Trained several students in Arduino, Raspberry Pi, and Python, enhancing their technical skills.
- **Communication Development** : Developed my personality and communication skills through interactions with students and parents.

SKILLS

Python

C++

ROS

ROS2

Machine Learning

Deep Learning

Robot Process Automation

Embedded C

Natural Language Processing

CERTIFICATES

Certificate of Publication (IJARIIT)(03/2022 - Present)

Published a research paper in IJARIT on the topic "Artificial Reaching and Feeding Hand"

Intenship (AI Master Class)(10/2020 - 11/2020)

Worked real time in python programming to perform Imageprocessing, Machine learning and Deep learning applications

Internship (12/2019)

Learned about Robotic arms and Robots used in industries. Worked real time on various sensors used in industries.

Inplant Training (06/2019)

Hackathon (02/2019)

INTERESTS

Robotics

Automation

Cricket

LANGUAGES

English

Full Professional Proficiency

Malayalam

Native or Bilingual Proficiency

Hindi

Full Professional Proficiency

Tamil

Professional Working Proficiency

Japanese

JLPT N3 Certified